Activity-4

Fill in the blanks with appropriate cardinal numbers.

S.No.	$n(A)$	$n(B)$	$n(A \cup B)$	$n(A \cap B)$	$n(A-B)$	$n(B-A)$
1	30	45	65			
2	20		55	10		
3	50	65		25		
4	30	43	70			

Example 1.29

Let $A=\{b, d, e, g, h\}$ and $B=\{a, e, c, h\}$. Verify that $n(A-B) = n(A)-n(A \cap B)$.

Solution

$A = \{b, d, e, g, h\}, B = \{a, e, c, h\}$

$A - B = \{b, d, g\}$

$n(A-B) = 3$ ... (1)

$A \cap B = \{e, h\}$

$n(A \cap B) = 2$, $n(A) = 5$

$n(A) - n(A \cap B) = 5-2$

$= 3$ ... (2)

Form (1) and (2) we get $n(A-B) = n(A)-n(A \cap B)$.

Example 1.30

In a school, all students play either Hockey or Cricket or both. 300 play Hockey, 250 play Cricket and 110 play both games. Find

(i) the number of students who play only Hockey.

(ii) the number of students who play only Cricket.

(iii) the total number of students in the School.

Solution:

Let H be the set of all students who play Hockey and C be the set of all students who play Cricket.

Then $n(H) = 300$, $n(C) = 250$ and $n(H \cap C) = 110$

Using Venn diagram,

From the Venn diagram,

(i) The number of students who play only Hockey = 190

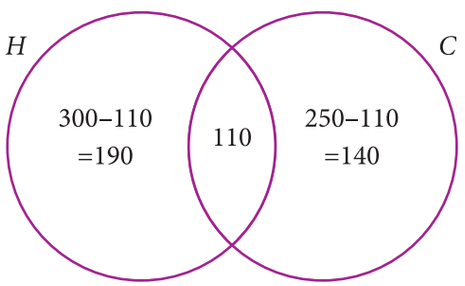


Fig. 1.34

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